

Quantum Active Inference: Topological Semantic Communication

The preceding sections have established that categorical string diagrams—from DisCoCat's pregroup derivations (??) through enriched hom-values (??) to topos-theoretic transfer (??)—provide a unified diagrammatic language for case-theoretic reasoning. This section extends the framework into its natural quantum generalization: topological quantum neural networks (TQNNs), the ZX-calculus, and sheaf-theoretic quantum semantic communication. The central observation is that the same monoidal-categorical architecture that underlies DisCoCat string diagrams also underlies quantum circuits, TQFT cobordisms, and quantum information flow—making active inference on case-marked relational structure a quantum topological computation.

Topological Quantum Neural Networks: Spin-Network Representations

From Quantum Neural Networks to TQFT

Marcianò, Fields, and Glazebrook demonstrate that quantum